

REVIEW

After the last Ozempic shot, your body changes on two clocks

REFERENCES	STYLE	TOKENS	VERIFICATION	COMPILED
31 verified	Popsci	≈6.4k	Crossref	28 Aug 2026

Ozempic does not vanish overnight, and your body does not snap into a chemical crash. The drug fades over weeks; appetite, weight and blood sugar answer over months, often moving back toward their old levels by different amounts.

In 2021, Domenica Rubino and the STEP 4 investigators divided more than eight hundred adults who had already spent five months taking semaglutide. Some continued the weekly obesity dose. The rest received placebo injections, while both groups kept the same lifestyle support. The volunteers who stayed on the drug kept losing weight; the other curve bent upward.¹

That split is the cleanest close-up we have of what follows a final dose. It is also narrower than the brand name in the question. STEP tested the high dose sold for obesity as Wegovy; Ozempic contains the same molecule but is usually prescribed at lower doses for type 2 diabetes. Across those uses, two clocks start after treatment ends: the medicine leaves slowly, then the body systems it had changed respond on their own schedules.^{1,2,3}

01 The first clock is chemical

Rune V. Overgaard and colleagues combined semaglutide studies that delivered the molecule by mouth, under the skin and into a vein. However it entered, the amount circulating in blood fell by half in about one week.⁴ After your final weekly injection, meaningful exposure therefore lingers for several weeks. There is no pharmacological cliff at midnight on day seven.

Semaglutide belongs to a class called a glucagon-like peptide-1 receptor agonist, a medicine that copies part of a gut hormone's signal to the brain and pancreas. As the concentration falls, that borrowed signal weakens. Reviews led by Damhof and John Blundell describe a return of the biology that treatment had been suppressing, rather than a separate dependence syndrome.^{5,6}

Some changes can feel welcome. Sean Wharton and the STEP safety team found that nausea, diarrhea and vomiting clustered around dose escalation and usually passed. After the randomized switch, new stomach or bowel symptoms appeared less often among those moved to placebo than among those who stayed on semaglutide.⁷ Areesha Moiz and Bibek Saha, reviewing the wider drug class, reached the same broad conclusion while warning that constipation can linger and persistent pain needs another explanation.^{8,9}

02 Hunger hears the fading signal

John Blundell's team gave adults with obesity free access to meals after twelve weeks on injectable semaglutide or placebo. On the medicine, volunteers ate about a quarter less over the day and reported fewer cravings.⁶ Martin Friedrichsen's group later used the obesity dose and watched participants eat roughly a third less at lunch.¹⁰

Gibbons and Gabe found similar appetite effects with oral semaglutide in people with diabetes or obesity.^{11,12} None of these teams followed every meal after the final dose. They cannot tell you which breakfast will feel different. But they do reveal the job the fading signal had been doing: when it weakens, hunger and the pull of food have room to return.⁵

The stomach itself does not explain the whole effect. After longer treatment, Friedrichsen's and Gabe's groups found no clear continuing delay in their indirect tests of stomach emptying even though people still ate less.^{10,12} Short-term slowing can happen, especially while doses rise, but the durable appetite effect reaches beyond food simply sitting in the stomach.^{9,7}

03 Then the scale begins to answer

Rubino's trial caught weight turning upward within months of the placebo switch.¹ John P. H. Wilding and the STEP 1 extension team followed a different group for a full year after both semaglutide and structured lifestyle treatment ended. On average, those volunteers regained about two-thirds of what they had lost. They still finished lighter than they had begun.²

The wider literature follows the same direction. Sara Berg and colleagues pooled randomized withdrawal studies and found that drugs producing larger losses generally left more weight available to regain.¹³ Reviews led by Patel and Kolli also found substantial rebound, although the studies mixed drugs, doses and follow-up schedules.^{14,15}

Sam West's team then tried to see beyond the edge of those trials. Their model projected that people stopping the newest incretin medicines could return to baseline around a year and a half later.¹⁶ Yet the strongest observations ended at about one year. Past that point, the clock face is drawn by a model.



Trace both clocks in [Figure 1](#). The blue lanes carry observations; the dashed orange lane marks the appetite step that feeding studies make plausible but withdrawal trials have not timed directly.

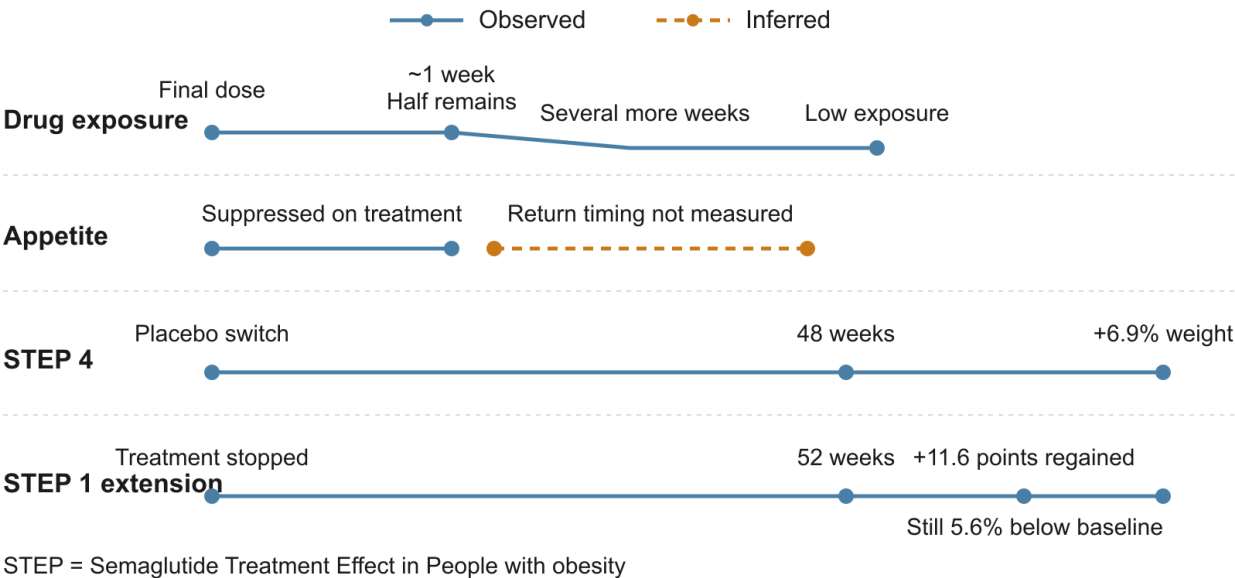


Figure 1. The drug leaves first; the measurable rebound takes months. Follow the lanes from the final dose toward the two weight studies. Blue means researchers observed the change; orange dashes mark the unmeasured timing between weaker appetite suppression and returning hunger. The trials remain separate because they enrolled different groups and ended treatment differently.^{4,6,1,2}

04 Blood sugar keeps another time

Chih-Chen Tzang and colleagues gathered eighteen randomized trials in which people stopped a medicine from the same broad family. In obesity trials, weight, waist size, blood pressure and glucose measures all moved in the less favorable direction. In the diabetes trials, weight return was smaller but HbA1c — a measure of average blood sugar over roughly three months — rose more.³

If you used Ozempic for diabetes, the scale can therefore miss the change that matters most. Other glucose-lowering medicines may cover part of the gap, which helps explain why fasting glucose did not rise consistently in the pooled trials.³ Jersey Jair Cárdenas-Salas and colleagues also showed in routine diabetes care that switching between active treatments can change both weight and glucose, making a clean stopping effect harder to isolate outside a trial.¹⁷

Wilding's extension showed the same uneven retreat among participants who began with prediabetes. Most had normal-range blood sugar when treatment ended; fewer than half did a year later, though the group still retained a small advantage over placebo.² McGowan and the STEP 10 investigators likewise watched glucose status improve during active treatment before a planned off-treatment phase.¹⁸ The body does not return every dial to its starting mark together.

The studies place their sharpest numbers here:

STUDY	PEOPLE	FOLLOW-UP AFTER SWITCH OR STOP	MAIN OBSERVED CHANGE	SOURCE
STEP 4	803 with obesity	48 weeks	+6.9% weight after placebo switch	¹
STEP 1 extension	327 with obesity	52 weeks	+11.6 percentage points regained; 5.6% below baseline	²
Obesity-trial pool	2,873	12–52 weeks	+5.63 kg; HbA1c +0.25 points	³
Diabetes-trial pool	853	varied	+2.03 kg; HbA1c +0.65 points	³
Health-system co-hort	7,881 without diabetes	up to 12 months	Every stop group remained lighter than baseline	¹⁹

05 The scale hides which tissue returns

Rory J. McCrimmon's team used dual-energy X-ray absorptiometry, a scan that separates fat from lean body mass, in adults taking diabetes-dose semaglutide. During treatment, both compartments shrank, although more fat than lean tissue disappeared.²⁰

Reviews led by Paschalis Karakasis, John A. Batsis and Naseem Eisa agree that lean tissue makes up a meaningful minority of the weight lost with these medicines, but their estimates spread widely because the studies and scans differ.^{21,22,23} That

matters especially when age or frailty makes strength precious. Donna H. Ryan and Locatelli have argued that muscle preservation and resistance exercise deserve direct testing during incretin treatment.^{24,25}

What returns after stopping is harder to see. Akhila Bhandarkar and colleagues found too few repeated post-withdrawal scans to divide regained weight confidently into fat and lean tissue.²⁶ Your bathroom scale can record the rebound. It cannot tell researchers what rebuilt it.

06 One cohort refuses the average curve

In 2025, Hamlet Gasoyan and colleagues opened the health records of nearly eight thousand people prescribed semaglutide or a related medicine for obesity. Those who stopped early, stopped later or stayed on treatment all remained lighter than baseline after a year. The recorded post-stop curves looked surprisingly steady.¹⁹

Could some bodies really settle there? The records cannot decide. Prescriptions do not reveal whether someone switched drugs, changed diet or began exercising, and clinic weights appear only when a person returns. Reimar W. Thomsen's review of routine care adds the same warning: real-world doses are often lower and stopping is common, but the reasons and next steps are poorly captured.²⁷

Randomization still gives the cleaner answer about cause. Rubino's semaglutide trial and Louis J. Aronne's tirzepatide withdrawal trial both held the surrounding program steady; average weight rose when active treatment was removed.^{1,28} Yet even Wilding's extension found that nearly half its participants still carried substantial weight loss a year later.² The average curve is real. It is not your forecast.

07 A better experiment starts at the final dose

Julie R. Lundgren and colleagues offer one clue from liraglutide, a shorter-acting member of the same drug family. During treatment, participants who combined the medicine with supervised exercise lost more weight and fat than those using either alone.²⁹ A year after everything stopped, Simon Birk Kjær Jensen's team found that the combination group kept more weight off than the group that had used liraglutide alone.³⁰ Their companion analysis suggests the two treatments had worked through different parts of daily life: the medicine changed appetite, while exercise changed movement and sitting time.³¹

That clue does not prove a semaglutide stopping plan. West's broader synthesis found no reliable average protection from behavioral support after weight-loss medicines ended, and Damhof's review found no prospective randomized test of abrupt semaglutide stopping against tapering or a lower maintenance dose.^{16,5}

Rubino's volunteers began at the same treatment point and then split into two groups. The experiment still missing would split the final dose three ways: stop, taper, or maintain a lower dose under identical support, while researchers follow appetite, blood sugar, strength and body composition for at least two years. Only then will the second clock tell us whether its pace can be changed.

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